1. Inverted Right Triangle Pattern

Program

```
#include <iostream>

using namespace std;

int main() {

   int n = 5;

   for(int i = n; i >= 1; i--) {

      for(int j = 1; j <= i; j++) {

         cout << "* ";

      }

      cout << endl;

   }

   return 0;

}
```

Output

```
#include <iostream>
using namespace std;
int main() {
    int n = 5;
    for(int i = n; i >= 1; i--) {
        for(int j = 1; j <= i; j++) {
            cout << "* ";
        }
        cout << endl;
    }
    return 0;
}
```

OUTPUT

```
* * * * *
* * * *
* * *
* *
*
```

2. Check Whether a Number is a Strong Number

Program

```
#include <iostream>

using namespace std;

int main() {

  int num = 145;

  int temp = num;

  int sum = 0;

  while(temp > 0) {

    int digit = temp % 10;

    int fact = 1;

    for(int i = 1; i <= digit; i++)

      fact = fact * i;

    sum = sum + fact;

    temp = temp / 10;

  }

  if(sum == num)

    cout << num << " is a Strong Number";

  else

    cout << num << " is not a Strong Number";

  return 0;

}
```

Output

```cpp
#include <iostream>
using namespace std;
int main() {
    int num = 145;
    int temp = num;
    int sum = 0;
    while(temp > 0) {
        int digit = temp % 10;
        int fact = 1;
        for(int i = 1; i <= digit; i++)
            fact = fact * i;
        sum = sum + fact;
        temp = temp / 10;
    }
    if(sum == num)
        cout << num << " is a Strong Number";
    else
        cout << num << " is not a Strong Number";
    return 0;
}
```

OUTPUT

```
145 is a Strong Number
```

3. Count the Frequency of Each Digit in a Number

Program

```cpp
#include <iostream>

using namespace std;

int main() {

  int num = 122333;

  int a[10] = {0};

  while(num > 0) {

    int d = num % 10;

    a[d]++;

    num = num / 10;

  }

  for(int i = 0; i < 10; i++) {

    if(a[i] > 0)

      cout << i << " = " << a[i] << endl;
```

}

return 0;

}

Output

```cpp
#include <iostream>
using namespace std;
int main() {
    int num = 122333;
    int a[10] = {0};
    while(num > 0) {
        int d = num % 10;
        a[d]++;
        num = num / 10;
    }
    for(int i = 0; i < 10; i++) {
        if(a[i] > 0)
            cout << i << " = " << a[i] << endl;
    }
    return 0;
}
```

OUTPUT

```
1 = 1
2 = 2
3 = 3
```

4. Count the Number of Divisors of a Number

Program

#include <iostream>

using namespace std;

int main() {

int num = 12;

int count = 0;

for(int i = 1; i <= num; i++) {

if(num % i == 0)

```
            count++;
    }
    cout << "Number of Divisors = " << count;
    return 0;
}
```

Output

```
#include <iostream>
using namespace std;
int main() {
    int num = 12;
    int count = 0;
    for(int i = 1; i <= num; i++) {
        if(num % i == 0)
            count++;
    }
    cout << "Number of Divisors = " << count;
    return 0;
}
```

OUTPUT

```
Number of Divisors = 6
```

5. Calculate Compound Interest

Program

```
#include <iostream>
#include <cmath>
using namespace std;
int main() {
    float p = 1000;
    float r = 10;
    float t = 2
```

```
    float amount = p * pow(1 + r / 100, t);
    float ci = amount - p;
    cout << "Compound Interest = " << ci << endl;
    cout << "Total Amount = " << amount;
    return 0;
}
```

Output

```
#include <iostream>
#include <cmath>
using namespace std;
int main() {
    float p = 1000;
    float r = 10;
    float t = 2;
    float amount = p * pow(1 + r / 100, t);
    float ci = amount - p;
    cout << "Compound Interest = " << ci << endl;
    cout << "Total Amount = " << amount;
    return 0;
}
```

OUTPUT

```
Compound Interest = 210
Total Amount = 1210
```